Inheritance

Mark scheme and worked solutions for 60 questions: 30 multiple choice, 30 structured.

1 · Part 1

Core · Chromosomes, genes and proteins · [1]

Answer: B

Explanation

Chromosomes are made of DNA. Starch and cellulose are carbohydrates, and fat is a lipid; none is the chromosome's genetic material.

2 · Part 1

Core · Chromosomes, genes and proteins · [1]

Answer: D

Explanation

A gene is a length of DNA coding for a protein. It is not a cell, membrane or the protein it encodes.

3 · Part 1

Core · Chromosomes, genes and proteins · [1]

Answer: A

Explanation

Alleles are alternative forms of one gene. Ribosomes make proteins, gametes are reproductive cells, and enzymes are proteins with catalytic roles.

4 · Part 1

Core · Chromosomes, genes and proteins · [1]

Answer: C

Explanation

Human males normally have XY and females XX in body cells. A single X is a gamete complement, not the normal male body-cell pair.

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5 · Part 1 Core Chromosomes, genes and proteins [1]

Answer: D

Explanation

One X from each gamete gives XX. The sperm in this example contributes no Y and each normal gamete contributes one sex chromosome.

6 · Part 1 Extended Chromosomes, genes and proteins [1]

Answer: B

Explanation

The order of DNA bases determines the order of amino acids. Cell position and starch do not specify that sequence.

7 · Part 1 Extended Chromosomes, genes and proteins [1]

Answer: C

Explanation

Different amino-acid sequences can produce different folding and shapes. Identical function or active sites is not guaranteed.

8 · Part 1 Extended Chromosomes, genes and proteins [1]

Answer: A

Explanation

DNA controls production of proteins including carriers. A changed carrier may alter transport without removing DNA or changing every protein.

9 · Part 1 Extended Chromosomes, genes and proteins [1]

Answer: B

Explanation

mRNA copies genetic information and moves to the cytoplasm. Chromosomes normally remain in the nucleus during this process.

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10 · Part 1 Extended Chromosomes, genes and proteins [1]

Answer: C

Explanation

Ribosomes assemble amino acids in the order specified by mRNA. The other locations are not the site of protein assembly.

11 · Part 1 Extended Chromosomes, genes and proteins [1]

Answer: D

Explanation

Most body cells have the same genes but express different subsets. Specialisation does not normally require loss of all DNA or entirely different genomes.

12 · Part 1 Extended Chromosomes, genes and proteins [1]

Answer: A

Explanation

Haploid means one set of chromosomes. Diploid means two sets; neither term describes a protein or a pair of cells.

13 · Part 1 Extended Chromosomes, genes and proteins [1]

Answer: C

Explanation

Diploid nuclei contain two chromosome sets. This refers to chromosome sets, not a fixed total of one chromosome or amino acids.

14 · Part 1 Extended Chromosomes, genes and proteins [1]

Answer: A

Explanation

There are 23 pairs, giving 46 chromosomes. Twenty-three is the normal haploid number; 44 counts only the autosomes.

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15 · Part 1 Extended Chromosomes, genes and proteins [1]

Answer: B

Explanation

The mRNA base sequence specifies the amino-acid sequence. Oxygen availability can affect metabolism but does not encode the sequence.

16 · Part 1 Core Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- Chromosomes are made of DNA. (1)
- A gene is a length of DNA that codes for a protein. (1)

(b) **[2]**

- An allele is an alternative form of a gene. (1)
- Different alleles have differences in their DNA sequence. (1)

> **ALLOW** alleles are different versions / forms of the same gene; alleles have different DNA

Worked solution

(a) Chromosomes are made of DNA. A gene is a length of DNA that codes for a protein.

(b) An allele is an alternative form of a gene. Different alleles have differences in their DNA sequence.

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17 · Part 1 Core Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- The mother has XX. (1)
- The father has XY. (1)

(b) **[2]**

- Eggs each contain an X chromosome. (1)
- Sperm contain either X or Y, giving XX or XY after fertilisation. (1)

ALLOW equal chance / 1 : 1 ratio of XX to XY

Worked solution

(a) Body cells are diploid, so each parent has a pair of sex chromosomes: the mother has XX and the father has XY.

(b) Gametes are haploid and carry only one sex chromosome. Every egg carries an X, because the mother has only X chromosomes to give. Half of the sperm carry an X and half carry a Y. The sex of the child therefore depends only on which sperm fertilises the egg.

	X (egg)
X (sperm)	XX (female)
Y (sperm)	XY (male)

The two outcomes XX and XY are produced in a 1 : 1 ratio, so each pregnancy has an equal chance of a female or a male child.

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18 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- The DNA base sequence differs between the alleles. (1)
- The base sequence determines the order of amino acids in the protein. (1)

(b) **[2]**

- A changed amino-acid sequence can alter the protein's shape. (1)
- A changed active-site shape may reduce binding of the usual substrate. (1)

Worked solution

(a) The DNA base sequence differs between the alleles. The base sequence determines the order of amino acids in the protein.

(b) A changed amino-acid sequence can alter the protein's shape. A changed active-site shape may reduce binding of the usual substrate.

19 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- The gene controls production of a protein, the enzyme. (1)
- The enzyme catalyses a reaction, so its production affects that reaction. (1)

(b) **[2]**

- Membrane carrier proteins. (1)
- Receptors for neurotransmitters. (1)

Worked solution

(a) The gene controls production of a protein, the enzyme. The enzyme catalyses a reaction, so its production affects that reaction.

(b) Membrane carrier proteins. Receptors for neurotransmitters.

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20 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- An mRNA copy of the gene is made in the nucleus. (1)
- The mRNA moves from the nucleus to the cytoplasm. (1)

(b) **[2]**

- The mRNA passes through a ribosome. (1)
- The ribosome assembles amino acids in the sequence specified by the mRNA. (1)

Worked solution

(a) An mRNA copy of the gene is made in the nucleus. The mRNA moves from the nucleus to the cytoplasm.

(b) The mRNA passes through a ribosome. The ribosome assembles amino acids in the sequence specified by the mRNA.

21 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- Most body cells in one individual contain the same genes. (1)
- A gene can be present without being expressed. (1)

(b) **[2]**

- Different cells express different genes. (1)
- They produce the particular proteins needed for their specific functions. (1)

Worked solution

(a) Most body cells in one individual contain the same genes. A gene can be present without being expressed.

(b) Different cells express different genes. They produce the particular proteins needed for their specific functions.

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22 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- Diploid means containing two sets of chromosomes. (1)
- The haploid number is 8. (1)

(b) **[2]**

- A haploid nucleus contains one set of chromosomes. (1)
- One set is half the diploid number, so 16 divided by 2 is 8. (1)

Worked solution

(a) Diploid means containing two sets of chromosomes. The haploid number is 8.

(b) A haploid nucleus contains one set of chromosomes. One set is half the diploid number, so 16 divided by 2 is 8.

23 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- A body-cell nucleus contains 46 chromosomes. (1)
- A gamete nucleus contains 23 chromosomes. (1)

(b) **[2]**

- The body-cell nucleus is diploid with two sets. (1)
- The gamete nucleus is haploid with one set. (1)

Worked solution

(a) A body-cell nucleus contains 46 chromosomes. A gamete nucleus contains 23 chromosomes.

(b) The body-cell nucleus is diploid with two sets. The gamete nucleus is haploid with one set.

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24 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- The gene remains in the nucleus. (1)
- An mRNA copy moves to the cytoplasm instead. (1)

(b) **[2]**

- A ribosome assembles amino acids to form the protein. (1)
- The sequence is specified by the mRNA rather than DNA being converted directly into protein. (1)

Worked solution

(a) The gene remains in the nucleus. An mRNA copy moves to the cytoplasm instead.

(b) A ribosome assembles amino acids to form the protein. The sequence is specified by the mRNA rather than DNA being converted directly into protein.

25 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- DNA controls production of the receptor protein. (1)
- Whether the receptor is present affects the cell's ability to detect the signal. (1)

(b) **[2]**

- The receptor gene is expressed in one cell type. (1)
- It is not expressed, or is expressed much less, in the other cell type. (1)

Worked solution

(a) DNA controls production of the receptor protein. Whether the receptor is present affects the cell's ability to detect the signal.

(b) The receptor gene is expressed in one cell type. It is not expressed, or is expressed much less, in the other cell type.

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26 · Part 1 Core Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- Some sperm contain X. (1)
- Other sperm contain Y. (1)

(b) **[2]**

- Fertilisation by an X-bearing or Y-bearing sperm is a new event. (1)
- Each outcome remains approximately equally likely regardless of earlier children. (1)

ALLOW 50% / 1 in 2 / 1 : 1 chance of XY for every child

Worked solution

(a) The father is XY, so when his gametes are made each sperm receives either the X or the Y chromosome. Half of the sperm carry X and half carry Y.

(b) Every egg carries an X. Whether a child is XX or XY depends only on which type of sperm fertilises that egg:

	X (egg)
X (sperm)	XX
Y (sperm)	XY

For every fertilisation the chance of XY is 1 in 2 (50%). Each fertilisation is a separate random event; the sperm "do not remember" the earlier children, so two XX children do not change the odds for the next one. An XY child is as likely as before, but it is never certain.

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27 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- The sequence of bases in the gene specifies amino-acid order. (1)
- The two individuals may have different alleles of that gene. (1)

(b) **[2]**

- Different amino-acid sequences can produce different protein shapes. (1)
- A change in shape may affect binding or transport of the usual substance. (1)

Worked solution

(a) The sequence of bases in the gene specifies amino-acid order. The two individuals may have different alleles of that gene.

(b) Different amino-acid sequences can produce different protein shapes. A change in shape may affect binding or transport of the usual substance.

28 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- The amino acids could be assembled in different sequences. (1)
- The correct order must be specified by genetic information. (1)

(b) **[2]**

- The mRNA base sequence supplies the sequence information. (1)
- The ribosome reads that information while assembling the amino acids. (1)

Worked solution

(a) The amino acids could be assembled in different sequences. The correct order must be specified by genetic information.

(b) The mRNA base sequence supplies the sequence information. The ribosome reads that information while assembling the amino acids.

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29 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- P has 10 divided by 2, giving 5. (1)
- Q has 28 divided by 2, giving 14. (1)

(b) **[2]**

- Diploid describes two complete sets, regardless of the number in each set. (1)
- Both species have chromosome pairs, although the number of pairs differs. (1)

Worked solution

(a) P has 10 divided by 2, giving 5. Q has 28 divided by 2, giving 14.

(b) Diploid describes two complete sets, regardless of the number in each set. Both species have chromosome pairs, although the number of pairs differs.

30 · Part 1 Extended Chromosomes, genes and proteins [4]

Mark scheme

(a) **[2]**

- A chromosome is made of DNA. (1)
- A gene is a length of that DNA coding for a protein. (1)

(b) **[2]**

- The changed gene can alter production of the enzyme protein. (1)
- Less enzyme may reduce the rate of the metabolic reaction it catalyses. (1)

Worked solution

(a) A chromosome is made of DNA. A gene is a length of that DNA coding for a protein.

(b) The changed gene can alter production of the enzyme protein. Less enzyme may reduce the rate of the metabolic reaction it catalyses.

1 · Part 2 Extended Mitosis [1]

Answer: B

Explanation

Mitosis gives genetically identical nuclei and maintains chromosome number. Halving is associated with meiosis and fusion with fertilisation.

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2 · Part 2 Extended Mitosis [1]

Answer: D

Explanation

Exact replication provides copies that can be separated into daughter nuclei. Destruction, fertilisation and reduction division are different processes.

3 · Part 2 Extended Mitosis [1]

Answer: A

Explanation

Stem cells divide by mitosis and their daughter cells can specialise. They contain chromosomes and are not defined by pollen production.

4 · Part 2 Extended Meiosis [1]

Answer: C

Explanation

Meiosis produces gametes with reduced chromosome number. Fertilisation joins gametes, while pollination and digestion have other roles.

5 · Part 2 Extended Meiosis [1]

Answer: D

Explanation

Meiosis halves 32 to 16 and produces genetically different cells. Neither chromosome doubling nor repeated halving to eight describes the process.

6 · Part 2 Core Monohybrid inheritance [1]

Answer: B

Explanation

Phenotype means observable features. Rr is the genotype; R and r are alleles, and a gamete is a reproductive cell.

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7 · Part 2 Core Monohybrid inheritance [1]

Answer: C

Explanation

Heterozygous means having two different alleles of a particular gene. Two identical alleles (dominant or recessive) describe homozygous genotypes, and a diploid individual carries two alleles of each gene, not one.

8 · Part 2 Core Monohybrid inheritance [1]

Answer: A

Explanation

The recessive feature is expressed only when no dominant allele is present, as in bb. BB and Bb both contain the dominant B, which is expressed whenever present, so Bb cannot show the recessive feature and option D is wrong.

9 · Part 2 Core Monohybrid inheritance [1]

Answer: B

Explanation

Pp produces P and p gametes; pp produces p gametes only. The offspring are equally likely Pp or pp, giving 1 : 1.

10 · Part 2 Core Monohybrid inheritance [1]

Answer: C

Explanation

A 3 : 1 ratio under complete dominance includes one homozygous recessive combination among four equally likely combinations. Both other genotypes express the dominant feature.

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11 · Part 2 Extended Monohybrid inheritance [1]

Answer: D

Explanation

A recessive partner allows recessive offspring to reveal a recessive allele in the unknown parent. A dominant partner can mask that distinction.

12 · Part 2 Extended Monohybrid inheritance [1]

Answer: A

Explanation

In codominance both alleles contribute to the heterozygote's phenotype. Neither allele is simply absent or completely masked.

13 · Part 2 Extended Monohybrid inheritance [1]

Answer: C

Explanation

Group O requires two I^o alleles. I^A and I^B are each dominant over I^o, and together they give group AB.

14 · Part 2 Extended Monohybrid inheritance [1]

Answer: A

Explanation

A male's single X can express a recessive colour-blindness allele without another X carrying a dominant allele to mask it.

15 · Part 2 Core Monohybrid inheritance [1]

Answer: B

Explanation

Two aa parents produce only a gametes, so all offspring are aa. Crosses containing heterozygotes can produce more than one genotype.

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16 · Part 2 Extended Mitosis [4]

Mark scheme

(a) **[2]**

- Each daughter nucleus has 24 chromosomes. (1)
- Copies of each replicated chromosome separate so each nucleus receives a complete set. (1)

ALLOW chromosomes are exactly replicated / copied before mitosis

(b) **[2]**

- growth (1)
- replacement of cells / asexual reproduction (1)

ALLOW any two of growth, replacement of cells, asexual reproduction

REJECT repair of damaged tissue (given in the question)

Worked solution

(a) Each daughter nucleus has 24 chromosomes. Copies of each replicated chromosome separate so each nucleus receives a complete set.

(b) Growth by increasing cell number. Asexual reproduction or replacement of worn-out cells.

17 · Part 2 Extended Mitosis [4]

Mark scheme

(a) **[2]**

- Each daughter nucleus needs a complete set of genetic information. (1)
- Exact copying helps both daughter nuclei receive the same information as the parent. (1)

(b) **[2]**

- The copies separate. (1)
- Each daughter nucleus receives one copy of each chromosome. (1)

Worked solution

(a) Each daughter nucleus needs a complete set of genetic information. Exact copying helps both daughter nuclei receive the same information as the parent.

(b) The copies separate. Each daughter nucleus receives one copy of each chromosome.

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18 · Part 2 Extended Mitosis [4]

Mark scheme

(a) **[2]**

- Stem cells are unspecialised cells. (1)
- They divide by mitosis to produce daughter cells. (1)

ALLOW their daughter cells can become specialised for specific functions

(b) **[2]**

- The daughter cells become specialised. (1)
- They develop the features needed for that tissue's particular function. (1)

Worked solution

(a) Stem cells are unspecialised cells. They divide by mitosis to produce daughter cells.

(b) The daughter cells become specialised. They develop the features needed for that tissue's particular function.

19 · Part 2 Extended Meiosis [4]

Mark scheme

(a) **[2]**

- Each gamete nucleus has 6 chromosomes. (1)
- Gametes are genetically different from one another. (1)

(b) **[2]**

- It halves the number of chromosome sets. (1)

ALLOW 12 chromosomes reduced to 6

- It changes a diploid nucleus to haploid nuclei. (1)

Worked solution

(a) Each gamete nucleus has 6 chromosomes. Gametes are genetically different from one another.

(b) It halves the number of chromosome sets. It changes a diploid nucleus to haploid nuclei.

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20 · Part 2 Core Monohybrid inheritance [4]

Mark scheme

(a) **[2]**

- Hh is its genotype, describing the alleles present. (1)
- Hairy leaves are its phenotype, an observable feature. (1)

(b) **[2]**

- Inheritance is transmission of genetic information from generation to generation. (1)
- Alleles of genes are passed from parents to offspring. (1)

Worked solution

(a) Hh is its genotype, describing the alleles present. Hairy leaves are its phenotype, an observable feature.

(b) Inheritance is transmission of genetic information from generation to generation. Alleles of genes are passed from parents to offspring.

21 · Part 2 Core Monohybrid inheritance [4]

Mark scheme

(a) **[2]**

- BB and bb are homozygous because each has two identical alleles. (1)
- Bb is heterozygous because its alleles differ. (1)

(b) **[2]**

- Bb is black because the dominant B allele is expressed. (1)
- bb is white because no dominant B allele is present. (1)

Worked solution

(a) BB and bb are homozygous because each has two identical alleles. Bb is heterozygous because its alleles differ.

(b) Bb is black because the dominant B allele is expressed. bb is white because no dominant B allele is present.

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22 · Part 2 Core Monohybrid inheritance [4]

Mark scheme

(a) **[2]**

- Each parent produces S and s gametes. (1)
- The offspring combinations are SS, Ss, Ss and ss. (1)

(b) **[2]**

- The ratio is 3 smooth : 1 wrinkled. (1)
- Only ss offspring are wrinkled because the dominant S allele is absent. (1)

Worked solution

(a) Each parent produces S and s gametes. The offspring combinations are SS, Ss, Ss and ss.

(b) The ratio is 3 smooth : 1 wrinkled. Only ss offspring are wrinkled because the dominant S allele is absent.

23 · Part 2 Core Monohybrid inheritance [4]

Mark scheme

(a) **[2]**

- The Ll parent produces L and l gametes. (1)
- The ll parent produces l gametes only. (1)

(b) **[2]**

- Offspring are equally likely Ll and ll. (1)
- The ratio is 1 long-winged : 1 short-winged. (1)

Worked solution

(a) The Ll parent produces L and l gametes. The ll parent produces l gametes only.

(b) Offspring are equally likely Ll and ll. The ratio is 1 long-winged : 1 short-winged.

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24 · Part 2 Extended Monohybrid inheritance [4]

Mark scheme

(a) **[2]**

- Cross the red-flowered plant with a white-flowered rr plant. (1)
- Record the flower colours of many offspring. (1)

(b) **[2]**

- White offspring show the unknown parent supplied r and is Rr. (1)
- A small sample could by chance contain only red offspring even if the parent is Rr. (1)

Worked solution

(a) Cross the red-flowered plant with a white-flowered rr plant. Record the flower colours of many offspring.

(b) White offspring show the unknown parent supplied r and is Rr. A small sample could by chance contain only red offspring even if the parent is Rr.

25 · Part 2 Extended Monohybrid inheritance [4]

Mark scheme

(a) **[2]**

- Both parents produce C^R and C^W gametes. (1)
- Offspring genotypes are C^RC^R, C^RC^W and C^WC^W. (1)

(b) **[2]**

- The ratio is 1 red : 2 mixed red-and-white : 1 white. (1)

ALLOW roan for mixed red-and-white

- Both alleles contribute to the heterozygote's phenotype because they are codominant. (1)

Worked solution

(a) Both parents produce C^R and C^W gametes. Offspring genotypes are C^RC^R, C^RC^W and C^WC^W.

(b) The ratio is 1 red : 2 mixed red-and-white : 1 white. Both alleles contribute to the heterozygote's phenotype because they are codominant.

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26 · Part 2 Extended Monohybrid inheritance [4]

Mark scheme

(a) **[2]**

- The first produces I^A and I^o gametes. (1)
- The second produces I^B and I^o gametes. (1)

(b) **[2]**

- The combinations give AB, A, B and O. (1)
- These groups occur in an expected ratio of 1 : 1 : 1 : 1. (1)

Worked solution

(a) The first produces I^A and I^o gametes. The second produces I^B and I^o gametes.

(b) The combinations give AB, A, B and O. These groups occur in an expected ratio of 1 : 1 : 1 : 1.

27 · Part 2 Extended Monohybrid inheritance [4]

Mark scheme

(a) **[2]**

- X^NX^N. (1)
- X^NX^n. (1)

(b) **[2]**

- Half the sons are expected to be X^nY. (1)
- A son receives Y from his father and has an equal chance of receiving either maternal X. (1)

Worked solution

(a) X^NX^N. X^NX^n.

(b) Half the sons are expected to be X^nY. A son receives Y from his father and has an equal chance of receiving either maternal X.

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28 · Part 2 Core Monohybrid inheritance [4]

Mark scheme

(a) **[2]**

- PP is consistent with the line. (1)
- Two PP parents pass on P only, so all offspring are PP and purple. (1)

(b) **[2]**

- Both parents can pass on p. (1)
- Some offspring will be pp and have white flowers. (1)

ALLOW one quarter / 3 : 1 purple : white

Worked solution

(a) PP is consistent with the line. Two PP parents pass on P only, so all offspring are PP and purple.

(b) Both parents can pass on p. Some offspring will be pp and have white flowers.

29 · Part 2 Extended Monohybrid inheritance [4]

Mark scheme

(a) **[2]**

- The AB parent is I^AI^B and produces I^A or I^B gametes. (1)
- The O parent is I^oI^o and produces I^o gametes only. (1)

(b) **[2]**

- Half are expected to be A and half B. (1)
- Each child receives I^o from the O parent, so no child receives both I^A and I^B. (1)

Worked solution

(a) The AB parent is I^AI^B and produces I^A or I^B gametes. The O parent is I^oI^o and produces I^o gametes only.

(b) Half are expected to be A and half B. Each child receives I^o from the O parent, so no child receives both I^A and I^B.

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30 · Part 2 Core Monohybrid inheritance [4]

Mark scheme

(a) **[2]**

- A heterozygote cross predicts three quarters dominant, so 200 × 3 / 4 = 150. (1)
- One quarter recessive gives 200 × 1 / 4 = 50. (1)

(b) **[2]**

- Which gametes fuse is a chance event. (1)
- An expected probability does not guarantee an exact ratio in a finite sample. (1)

Worked solution

(a) A heterozygote cross predicts three quarters dominant, so 200 × 3 / 4 = 150. One quarter recessive gives 200 × 1 / 4 = 50.

(b) Which gametes fuse is a chance event. An expected probability does not guarantee an exact ratio in a finite sample.

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